MODULE -2: AC Fundamentals

In practice commonly used appliances and equipment's in residential and industries are operated from Alternating Current (AC) supply. ***The use of AC supply is preferred over DC supply, because it is easier to generate, transform to higher voltages, and transmit over long distances with lesser power transmission losses.***

Important Terminologies : The following terminologies are commonly used in analysis of AC circuits. Consider an AC voltage shown below.

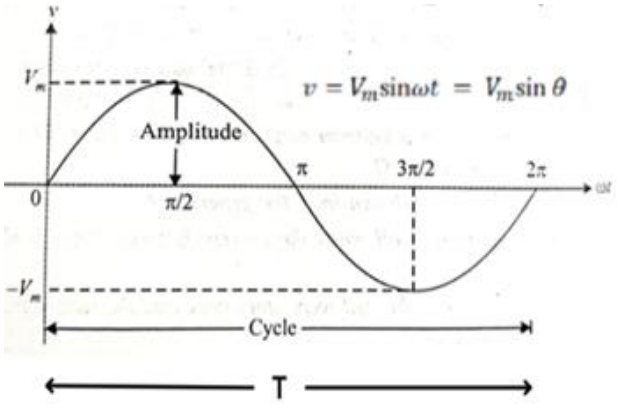


Fig. Sinusoidal Waveform of AC Voltage

1. **Cycle :** One complete set of variation of positive and negative values of an alternating quantity (voltage or current) is called a ***Cycle*****.**
2. **Amplitude or Maximum Value or Peak Value:** It is the maximum value of an alternating quantity (Voltage $\mathbf{V_m}$ or Current $\mathbf{I_m}$).
3. **Instantaneous Value:** It is the value of an alternating quantity varies with respect to time, Ex: Instantaneous voltage $\boldsymbol{v(t)}$ or Instantaneous current $\boldsymbol{i(t)}$ or Instantaneous Power $\boldsymbol{p(t)}$
4. **Time Period:** It is the time taken by an alternating quantity (Voltage or Current) to complete one full cycle of variation **0 to 360^0** or $\boldsymbol{2\pi}$ radians, and is denoted as **T** as shown above.
5. **Frequency :** It is defined as the number of cycles of variation of signal (voltage or current) in one second. It is denoted as **f** and its unit is Hertz (Hz) or Cycles Per Second (CPS). It is expressed as reciprocal of time period **T**.

$$f = \frac{1}{T}$$

The frequency of AC supply employed in **India is 50 Hz** and in **USA and UK is 60 Hz**.

If, f = 50 Hz, then its corresponding time period, T = 1/50 = 0.02 s = 20 ms

6. **Angular Frequency (ω):** The values of a sine function repeat after every 2π radians. Angular frequency, denoted as **ω,** is equal to the number of radians covered in one second.

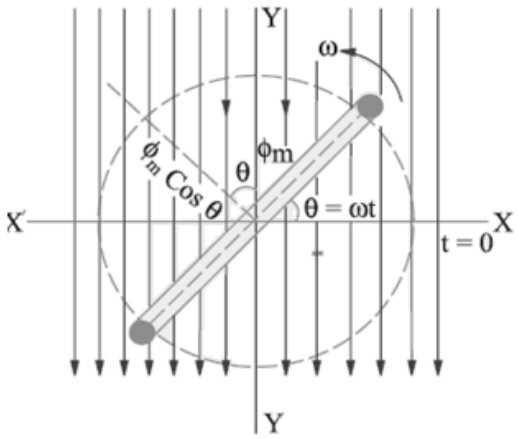
$$\omega = 2\pi f = \frac{2\pi}{T} \quad rad/s$$

7. **Phase :** It is the fraction of the time-period or cycle that has elapsed since it last passed from the chosen zero position or origin. The phase at time interval **t** from the selected origin is expressed as **t/T**.
8. **Phase Angle (θ) :** It is the equivalent of phase expressed in radians or degrees.

   Phase angle, **θ =2πt/T = 2πft = ωt**

   The maximum value of sinusoidal current occurs at a phase angle of **π/2** radians or **90°**.

Equation of AC Voltage and Current : Consider a rectangular coil, having *N* turns and rotating in a uniform magnetic field, with an angular velocity of *ω* radians/second, as shown in Fig.



Maximum flux $\emptyset_m$ is linked with the coil, when its plane coincides with the *X*-axis.

In a time interval of ***t*** seconds, this coil rotates through an angle $\theta = \omega t$.

At this deflected position, the component of the flux which is perpendicular to the plane of the coil, is $\emptyset = \emptyset_m \cos \omega t$.

Hence, **flux linkages** of the coil at any time is expressed as

$$\lambda = N\emptyset = N\emptyset_m \cos \omega t.$$

According to Faraday's Laws of Electromagnetic Induction,

$$e = -\frac{d}{dt}(N\Phi) \text{ volt} = -N.\frac{d}{dt}(\Phi_m \cos \omega t) \text{ volt} = -N\Phi_m\, \omega(-\sin \omega t) \text{ volt}$$

$$e = E_m \sin \omega t \qquad \text{.....(1)}$$

When the coil is rotated through 90^0, that means when $\omega t = \theta = 90^0$, then $\sin\theta = 1$, hence instantaneous emf has maximum value,

from the equation (1), $e = E_m$

where

$$E_m = \omega N \Phi_m = \omega N B_m A = 2\pi f N B_m A \text{ volt}$$

B_m = maximum value of magnetic flux density in wb/m^2 ;

A = Area of the coil in m^2

f = Frequency of rotation of the coil in cycles per second or Hz

Wave forms of AC Voltage and AC Current: A voltage which changes its polarity at regular intervals of time is called alternative voltage. The sinusoidal variation of the voltage is shown below:
The sinusoidal alternating voltage is expressed as follows.

$$v = V_m \sin\omega t = V_m \sin\theta$$

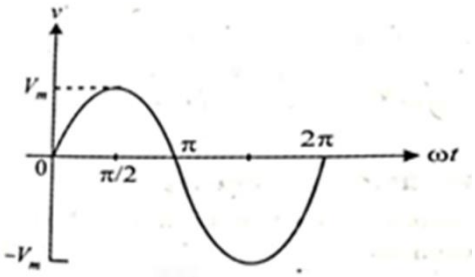
Where,

v = Instantaneous value of alternating voltage.

V_m = Maximum Value of Alternating Voltage.

ω = Angular velocity of the coil.

$\theta = Phase\ Angle = \omega t$



(i) Sinusoidal AC Voltage

When Sinusoidal voltage is applied to the circuit, a sinusoidal current flows through it. It is expressed as follows:

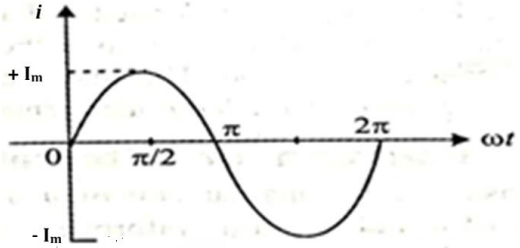
$$i = I_m \sin\omega t$$

Where,

i = Instantaneous value of alternating current.

I_m = Maximum Value of Alternating current.

ω = Angular velocity of the coil.



(ii) Sinusoidal AC current

Phase difference: A leading alternating quantity is one which reaches its maximum (or zero) value earlier as compared to the other quantity. The instantaneous emf 'v_B' leads the instantaneous emf 'v_A' by an angle 'ϕ' degrees as shown below.

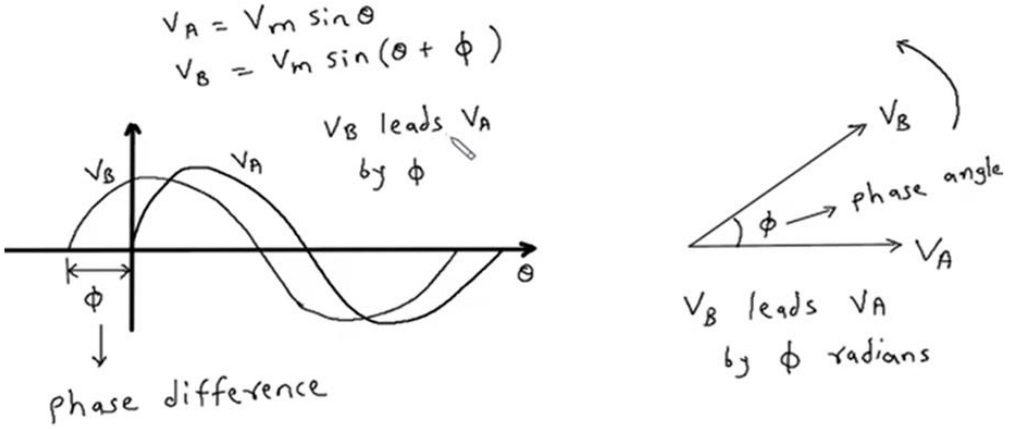


Fig. leading phase difference

Similarly, a lagging alternating quantity is one which reaches its maximum or zero value later than the other quantity. The instantaneous emf 'v_B' lags the instantaneous emf 'v_A' by an angle 'ϕ' degrees as shown below.

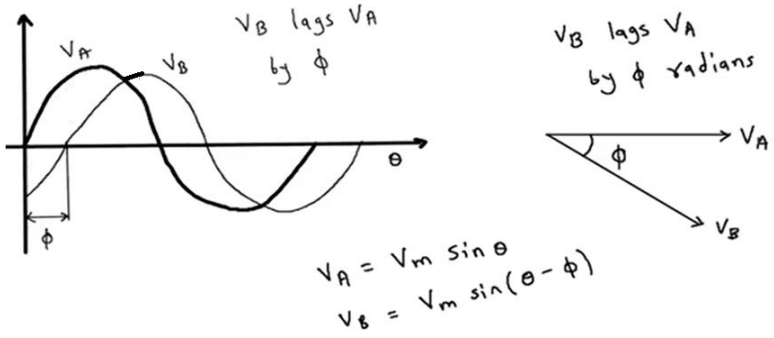
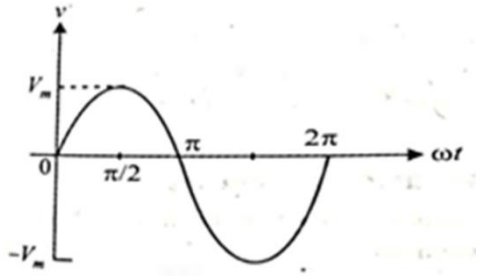


Fig. lagging phase difference

Average Value : It is defined as the algebraic sum of all values of an alternating quantity divided by the total number of values.

The average value of a sine wave over one full cycle is zero. Hence, average value of an alternating quantity for only half cycle is considered

$$\textbf{Average value} = \frac{\textbf{Area under one cycle}}{\textbf{Base length of one cycle}}$$



(i) Sinusoidal AC Voltage

Derivation:

$$V_{avg} = \frac{1}{\pi}\int_0^{\pi} v\, d\theta = \frac{1}{\pi}\int_0^{\pi} v\, d(\omega t)$$

$$= \frac{1}{\pi}\int_0^{\pi} V_m \sin\omega t\, d(\omega t) = \frac{V_m}{\pi}(-cos\omega t)_0^{\pi} = \frac{V_m}{\pi}(-cos\pi - (-\cos(o))$$

$$\therefore \quad \boldsymbol{V_{avg} = \frac{2\,V_m}{\pi} = 0.637 V_m}$$

Similarly, the average value of current is expressed as follows,

$$\boldsymbol{I_{avg} = \frac{2\,I_m}{\pi} = 0.637 I_m}$$

Root Mean Square (RMS or rms) Value : It is defined as a steady current (DC), when flowing through a given circuit for a given time produces the same heat as produced by the alternating current when flowing through the same circuit for the same time. It is also known as the **effective value** and represented as **I** or $\mathbf{I_{rms}}$.

The standard form of a sinusoidal alternating current is $\boldsymbol{i = I_m \sin \omega t = I_m \sin \theta}$.

The mean of the squares of the instantaneous values of current over one complete cycle is

$$\mathbf{I^2} = \int_0^{2\pi} \frac{i^2\, d\theta}{(2\pi - 0)}$$

The square root of this value is $\mathbf{I} = \sqrt{\left(\int_0^{2\pi} \frac{i^2\, d\theta}{2\pi}\right)}$

Hence, the r.m.s. value of the alternating current is

$$I = \sqrt{\left(\int_0^{2\pi} \frac{i^2 d\theta}{2\pi}\right)} = \sqrt{\left(\frac{I_m^2}{2\pi}\int_0^{2\pi} \sin^2\theta\, d\theta\right)} \qquad (\text{put } i = I_m \sin\theta)$$

Now, $\cos 2\theta = 1 - 2\sin^2\theta \quad \therefore \sin^2\theta = \frac{1-\cos 2\theta}{2}$

$$\therefore \qquad I = \sqrt{\left(\frac{I_m^2}{4\pi}\int_0^{2\pi} (1-\cos 2\theta)\, d\theta\right)} = \sqrt{\left(\frac{I_m^2}{4\pi}\left|\theta - \frac{\sin 2\theta}{2}\right|_0^{2\pi}\right)}$$

$$= \sqrt{\frac{I_m^2}{4}\ 2} \qquad \sqrt{\frac{I_m^2}{2}} \qquad \therefore \mathrm{I} = \frac{I_m}{\sqrt{2}} = 0.707\, I_m$$

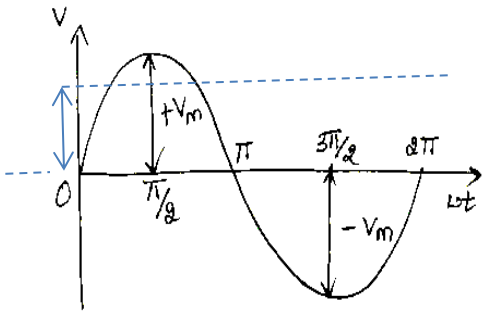
Hence, we find that for a symmetrical sinusoidal current

r.m.s. value of current = 0.707 × max. value of current

Hence, RMS value is also called as the square root of mean of square of different instantaneous values.

Similarly, RMS value of Voltage is expressed as, $V_{rms} = \frac{V_m}{\sqrt{2}} = 0.7071 V_m$

V = $V_m / \sqrt{2}$ = 0.707 V_m



Form Factor (FF): It is defined as the ratio of RMS value to the average value of an alternating quantity.

$$Form\ Factor = \frac{RMS\ Value}{Average\ Value}$$

For voltage wave,

$$Form\ Factor = \frac{V_{rms}}{V_{avg}} = \frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}} = \frac{0.707 V_m}{0.637 V_m} = 1.11$$

For current wave,

$$Form\ Factor = \frac{I_{rms}}{I_{avg}} = \frac{\frac{I_m}{\sqrt{2}}}{\frac{2I_m}{\pi}} = \frac{0.7071 I_m}{0.637 I_m} = 1.11$$

Peak Factor or Crest Factor or Amplitude Factor : The ratio of maximum value to the RMS value of an alternating quantity is known as the **peak factor.**

$$Peak\ Factor = \frac{Maximum\ Value}{RMS\ value}$$

For sinusoidal voltage wave ,

$$Peak\ Factor = \frac{V_m}{V_{rms}} = \frac{V_m}{\frac{V_m}{\sqrt{2}}} = \sqrt{2} = 1.414$$

For sinusoidal current wave,

$$Peak\ Factor = \frac{I_m}{I_{rms}} = \frac{I_m}{\frac{I_m}{\sqrt{2}}} = 1.414$$

Power Factor (pf or cosϕ) : It is defined as the cosine of phase angle between voltage and current. Power factor (pf) is a measure of how effectively electrical power is being used in an AC circuit.

If phase angle $\phi = 0^0$, $\cos\phi = 0$; If phase angle $\phi = 90^0$, $\cos\phi = 1$;

The value of power factor lies between 0 and 1.

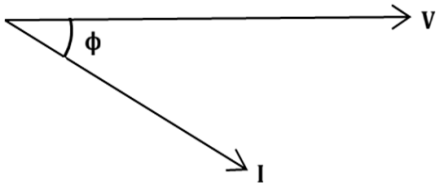


Fig : Phasor diagram for lagging pf load (inductive Load)

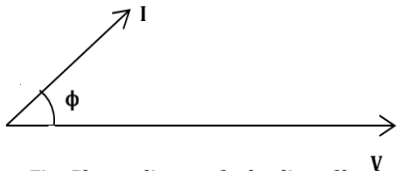


Fig : Phasor diagram for leading pf load (Capacitive Load)

Power in AC Circuits : It is defined as the rate of energy transfer or converted. The complex power **(S)** in AC circuits consists of three types of power and expressed as follows.

$$S = P + j\,Q$$

where

P = Real Power which is the actual power consumed by the load and is measured in watts (W);

Q = Reactive Power which is the power that cycles or oscillates between the source and reactive components like inductors or capacitors. It does not consume power, but required to form magnetic and electrical fields. Q = 0 for resistive loads. It is expressed in VAr (Volt Ampere Reactive)

S = Complex Power or Apparent Power represents overall power flow in the circuit and is expressed in Volt Amps (VA) or kVA or MVA.

Phasor representation of complex power using power triangle is as follows:

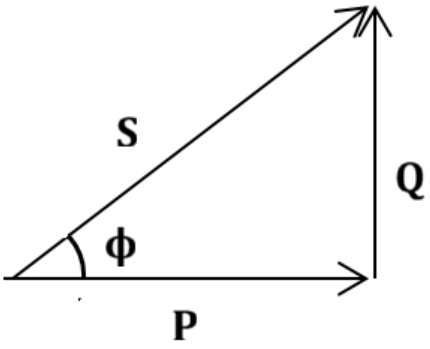


Fig : Power Triangle for inductive Loads

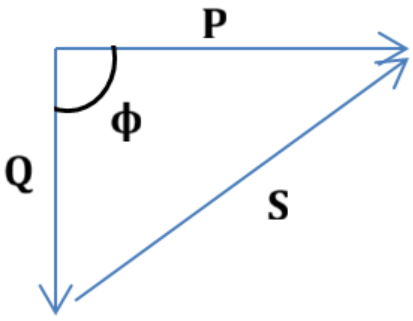


Fig : Power Triangle for Capacitive Loads

$$S = V \times I = \sqrt{P^2 + Q^2} \quad VA$$

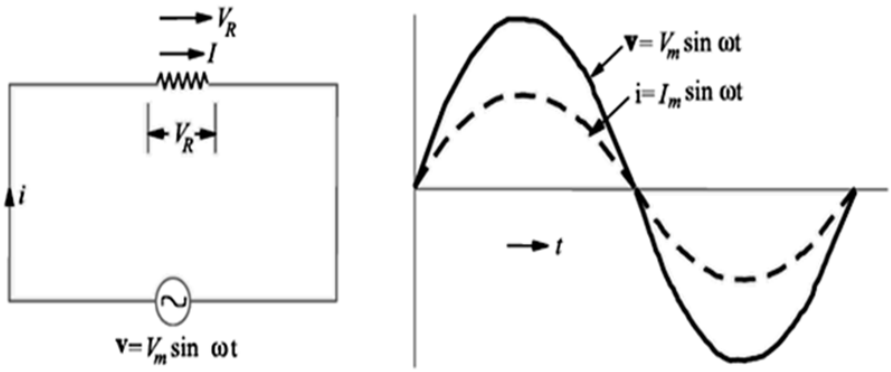
$$P = V\,I\cos\emptyset = S\cos\emptyset \quad \text{watts or W}$$

$$Q = V\,I\sin\emptyset = S\sin\emptyset \quad \text{VAr}$$

Series AC Circuits

An AC circuit is an electrical circuit powered by an alternating source. Series AC circuits consists of series connected elements like resistors, inductors, capacitors and one or more voltage sources.

Analysis of Purely Resistive Circuit :



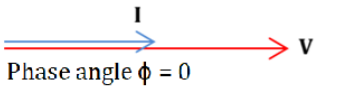


Fig : Phasor diagram of purely resistive AC circuit

Consider the circuit consisting of pure resistor with resistance of R ohms is connected across AC supply as shown above. The waveforms and phasor diagram of applied voltage applied and current flowing through the purely resistive AC circuit is also shown above.

Applied instantaneous voltage $\boldsymbol{v = V_m \sin \theta = V_m \sin \omega t}$ .....(1)

R = The resistance of the resistor in ohms.

$\boldsymbol{i}$ = $\boldsymbol{i}$(t) = instantaneous value of current in Amps.

By applying Ohm's law

$$\boldsymbol{i = v/R = \frac{Vm}{R} \sin \omega t = I_m \sin \omega t} \qquad \text{.....(2)}$$

where,

Maximum value of current, $I_m = V_m/R$,

RMS value of Current $\mathbf{I} = I_{rms} = \frac{I_m}{\sqrt{2}}$

Vector diagram or phasor diagram of purely resistive circuit is shown above. In this phasor diagram, Voltage and current are in phase with each other, and no phase difference exist between V and I.

Phase angle ϕ = 0

$\therefore$ $For\ purely\ resistive\ circuits, pf\ cos\emptyset = \cos(0) = 1$

From power triangle equations,

$$\mathbf{P = V\ I \cos\emptyset = V\ I \cos(0) = V\ I} \qquad \textbf{watts or W}$$

$$\mathbf{Q = V\ I \sin\emptyset = V\ I \sin(0) = 0} \qquad \textbf{VAr}$$

So reactive power Q is not present in purely resistive circuits, but only real power P is present. Hence, power supplied from the voltage source is absorbed or consumed by the resistance or the resistive load.

Multiplying the equations (1) and (2), we get

$$\text{Power. Instantaneous power, } p = vi = \mathbf{V_m \sin \omega t\ I_m \sin \omega t} = V_m I_m \sin^2 \omega t$$

$$= \frac{V_m I_m}{2}(1 - \cos 2\omega t) = \frac{V_m I_m}{2} - \frac{V_m I_m}{2} \cos 2\ \omega t$$

Power consists of a constant part $\frac{V_m I_m}{2}$ and a fluctuating part $\frac{V_m I_m \cos 2\omega t}{2}$ of frequency double that of voltage and current waves. Fluctuating part will be zero over a complete cycle.

$$P = \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}}$$

or

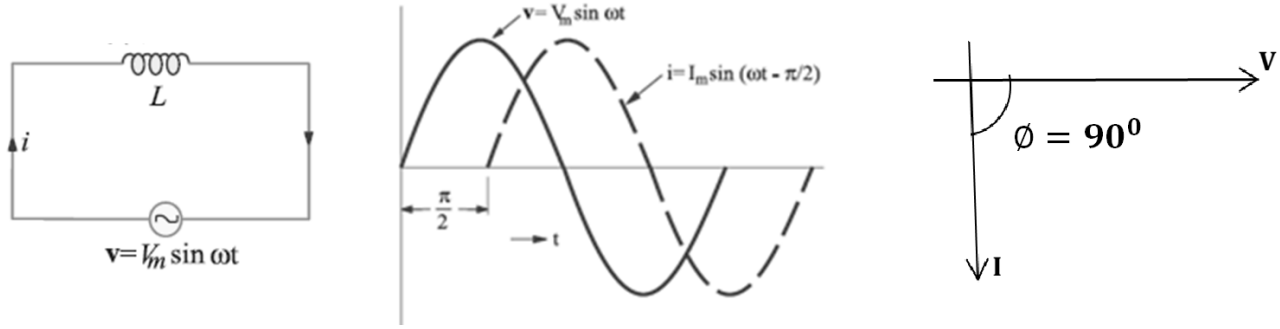
$$P = V \times I \text{ watts}$$

Where,

V = RMS value of applied voltage.

I = RMS value of the current flowing through the circuit.

Analysis of Purely Inductive Circuit : Consider the circuit consisting of pure inductor with inductance of **L** Henries is connected across AC supply as shown below. Whenever an alternating voltage is applied to a purely inductive coil, a back e.m.f. is produced due to the self-inductance of the coil and it opposes change of current through it.



$$v = L\frac{di}{dt}$$

$$v = V_m \sin \omega t$$

$$\therefore \quad V_m \sin \omega t = L\frac{di}{dt}; \qquad \therefore \; di = \frac{V_m}{L} \sin \omega t \, dt$$

Integrating both sides, we get $i = \frac{V_m}{L} \int \sin \omega t \, dt$

$$= \frac{V_m}{\omega L}(-\cos \omega t)$$

$$= \frac{V_m}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right) = \frac{V_m}{X_L} \sin(\omega t - \pi/2)$$

Max. value of i is $I_m = \frac{V_m}{\omega L}$ when $\sin\left(\omega t - \frac{\pi}{2}\right)$ is unity.

∴ Current lags applied voltage by $\pi/2$ or 90^0. The waveform and phasor diagram of voltage and current for purely inductive load is shown above.

The equation of the instantaneous current is given by

$$i = I_m \sin(\omega t - \pi/2).$$

$$I_m = V_m/\omega L = V_m/X_L.$$

where

$X_L = \omega L = 2\pi f L$ ohm.

X_L = Inductive **reactance** of the coil and its unit is ohms (Ω)

L = *Self-inductance of the coil* in henry or H

ω = Angular frequency in radians/second or rad / s.

Note : The inductive reactance capacitive reactance (X_L) varies directly with the frequency. For a steady current or DC current $\boldsymbol{f = 0}$, **hence $X_L = 0$ Ω.**

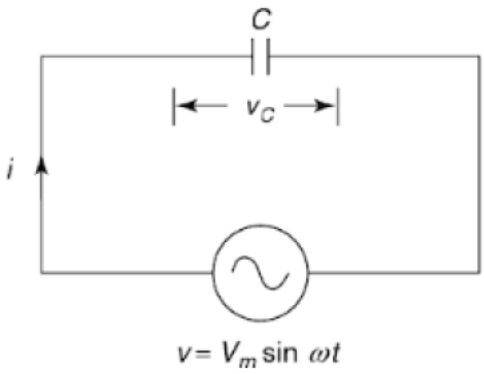
Power:

$$\text{Instantaneous power} = v_i = V_m I_m \sin\omega t \sin\left(\omega t - \frac{\pi}{2}\right) = V_m I_m \sin\omega t \cos\omega t = -\frac{V_m I_m}{2}\sin 2\omega t$$

$$\text{Power for whole cycle is } P = -\frac{V_m I_m}{2}\int_0^{2\pi} \sin 2\omega t \, dt = 0$$

The average power consumed for a complete cycle for pure inductor circuit is zero.

Analysis of Pure Capacitance in a A.C circuit : The capacitor is used to store electrical energy by accumulating electric charges between the two plates separated by an dielectric material like glass, paper, mica, oxide layers, etc. The ability of a capacitor to store electric charges or electrical energy is called **capacitance**, and is measured in Farads (F).



Consider a circuit consisting of a pure capacitor with a capacitance of **C** Farads is connected across an alternating voltage source as shown above.

$$\boldsymbol{v = V_m \sin(\omega t)} \qquad \ldots\ldots\ldots\ldots\ldots (1)$$

When an alternating voltage is applied to the capacitor, it is charged to a voltage of v_C. Let q be the instantaneous charge on the capacitor. Then voltage across the capacitor at that instant

$$\boldsymbol{v_c = \frac{q}{C}}$$

According to KVL,

$$\boldsymbol{v_c = v}$$

$$\boldsymbol{\frac{q}{C} = V_m \sin(\omega t)}$$

$$q = C V_m \sin \omega t$$

By the definition of current,

$$i = \frac{dq}{dt} = \frac{d}{dt}\left(CV_m \sin \omega t\right)$$

$$= CV_m \frac{d}{dt}\left(\sin \omega t\right)$$

$$= CV_m \omega \cos \omega t$$

$$\text{or} \quad i = \frac{V_m}{1/C\omega} \sin\left(\omega t + \pi/2\right)$$

Hence,

$$i = I_m \sin(\omega t + \pi/2)$$

∴ In pure capacitor AC circuit, the current leads the voltage by an angle of $\pi/2$ or 90^0

where

$$I_m = \frac{V_m}{\left(\frac{1}{\omega C}\right)} = \frac{V_m}{X_c} = \omega C V_m$$

$X_c = \left(\frac{1}{\omega C}\right) =$ **Capacitive reactance** in ohms

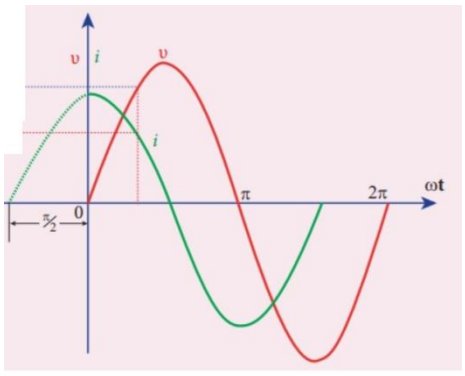
C = Capacitance of the Capacitor in Farads or F

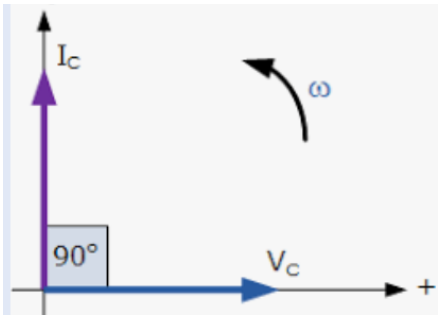
ω = Angular frequency in rad/s

The capacitive reactance (X_C) varies inversely as the frequency. For a steady current or DC current $f = 0$, hence ,

$$\therefore X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{0} = \infty$$

Thus a capacitive circuit offers infinite resistance to the steady current. So that steady current cannot flow through the capacitor.





The waveform and phasor diagram of voltage and current for purely inductive load is shown above.

Power: Instantaneous power

$$p = p(t) = vi$$

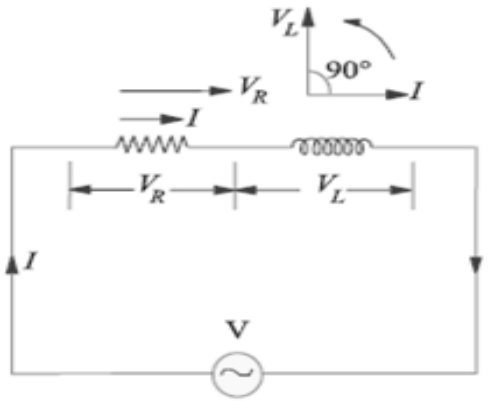
$$= V_m \sin \omega t . I_m \sin (\omega t + 90°)$$

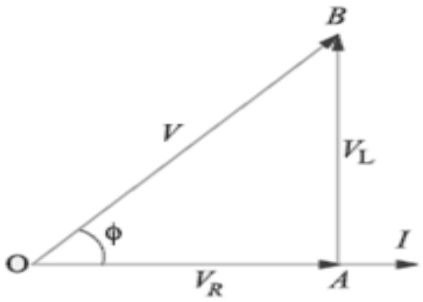
Power for the whole cycle

$$= \frac{1}{2} V_m I_m \int_0^{2\pi} \sin 2\omega t \, dt = 0$$

The average power consumed for a complete cycle for pure capacitor circuit is zero.

Analysis of R-L Series Circuit: A pure resistor with resistance of R ohms and a pure inductive coil of inductance ***L*** Henries are connected in series and connected across an AC supply as shown below.





Let,

V = r.m.s. value of the applied voltage,

I = r.m.s. value of the current.

Voltage drop across Resistor, $\mathbf{V_R = IR}$,

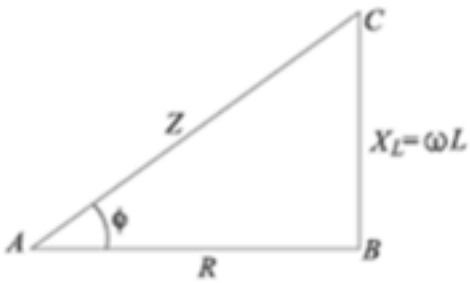
Voltage drop across Inductive Reactance, $\mathbf{V_L = IX_L}$

These voltage drops are shown in voltage triangle *OAB*. Vector *OA* represents ohmic drop V_R and vector *AB* represents inductive drop V_L. The applied voltage V is the vector sum of the two *i.e. OB*.

$$V = \sqrt{(V_R^2 + V_L^2)} = \sqrt{[(IR)^2 + (I . X_L)^2]} = I\sqrt{R^2 + X_L^2}$$

$$\mathbf{V = I Z}$$

The quantity $[\boldsymbol{Z}] = \boldsymbol{Z} = \sqrt{\boldsymbol{R^2 + X_L^2}}$ is known as the **impedance** (Z) of the circuit.



As seen from the impedance triangle *ABC*

$$Z^2 = R^2 + X_L^2.$$

(Impedance)² = (Resistance)² + (Reactance)²

From Fig. it is clear that the applied voltage V leads the current I by an angle φ such that

$$\tan\phi = \frac{V_L}{V_R} = \frac{I \cdot X_L}{I \cdot R} = \frac{X_L}{R} = \frac{\omega L}{R} = \frac{\text{reactance}}{\text{reactance}} \quad \therefore \phi = \tan^{-1}\frac{X_L}{R}$$

Current lags the applied voltage by an angle φ.

$$\mathbf{Z = R + jX_L}$$

Hence, if applied voltage is given by

$$\mathbf{v = V_m \sin \omega t,}$$

then current equation is

$$\mathbf{i = I_m \sin (\omega t - \phi),}$$

where

$$\mathbf{I_m = \frac{Vm}{Z}}$$

Power: Instantaneous power $p = v\,i$

$$= V_m \sin\omega t \times I_m \sin(\omega t - \phi) = V_m I_m \sin\omega t \sin(\omega t - \phi)$$

$$= \frac{1}{2} V_m I_m [\cos\phi - \cos(2\omega t - \Phi)]$$

The above equation consists of following terms :

a) **(1 /2)** $\boldsymbol{V_m I_m}$ **cos** ϕ - Constant part

b) **(1/2)**$\boldsymbol{V_m I_m}$ **cos (2ω** $\boldsymbol{t}$ **–** ϕ**)** - Pulsating component since its average value over a complete cycle is zero.

Average power consumed = (1/2) $\boldsymbol{V_m I_m}$ **cos ϕ = V I cos ϕ**

Where, V and I represent the r.m.s values of voltage and current.

Analysis of R C Series Circuit : A pure resistor with a resistance ***R*** *ohms* and a pure capacitor with a capacitance of **C** *Farads* are connected in series and connected across an AC supply as shown below.

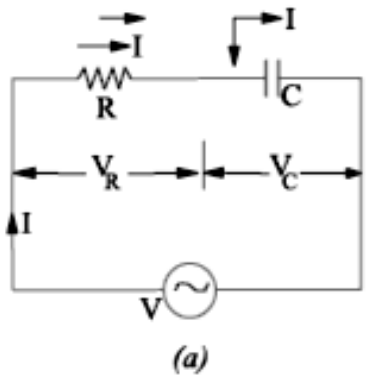
Let

V = r.m.s. value of the applied voltage,

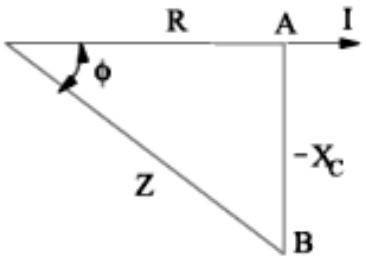
I = r.m.s. value of the current.

Voltage drop across Resistor $V_R = I\,R$,

Voltage drop across Capacitive Reactance $V_c = I\,Xc$



(a)

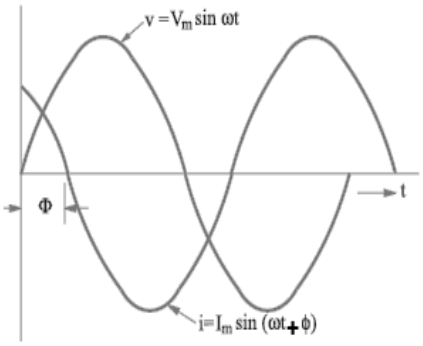


Impedance Triangle

$$V = \sqrt{V_R^2 + (-V_C)^2} = \sqrt{(IR)^2 + (-IX_C)^2} = I\sqrt{R^2 + X_C^2} \quad \text{or} \quad I = \frac{V}{\sqrt{R^2 + X_C^2}} = \frac{V}{Z}$$

The denominator is called the **impedance** of the circuit

$$Z = \sqrt{R^2 + X_C^2}$$



I leads **V** by angle ϕ such that

$$\tan \phi = -X_C/R$$

$$v = V_m \sin \omega t,$$

$$i = I_m \sin (\omega t + \phi)$$

so that current **leads** the applied voltage by an angle ϕ as shown above.

Power : Instantaneous power, $p = v\,i$

$$= V_m \sin\omega t \times I_m \sin(\omega t + \phi) = V_m I_m \sin\omega t \sin(\omega t + \phi)$$

$$= \frac{1}{2} V_m I_m [\cos\phi - \cos(2\omega t + \Phi)]$$

The above equation consists of following terms :

a) **(1 /2)** $V_m I_m$ **cos** ϕ - Constant part

b) **(1/2)**$V_m I_m$ **cos (2ω** t **+** ϕ **)** - Pulsating component since its average value over a complete cycle is zero.

Average power consumed = (1/2) $V_m I_m$ **cos ϕ = VI cos ϕ .**

where

V and I represent the r.m.s values of voltage and current.

Analysis of Series R-L-C Circuit

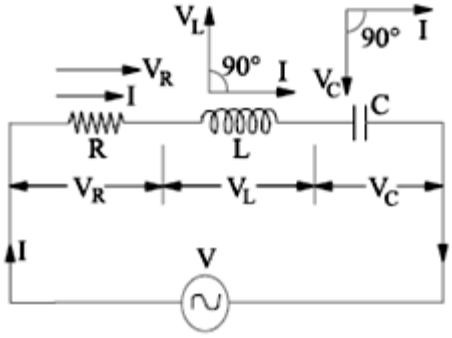
A pure resistor with resistance of **R** ohms, a pure Inductor with inductance of **L** Henries and a pure capacitor with capacitance of **C** Farads are connected in series and connected across an AC Supply as shown below.

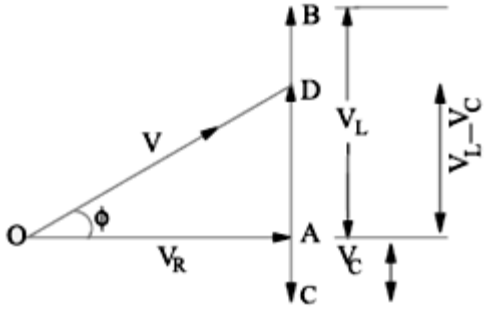
Let,

V_R = IR = voltage drop across R —in phase with I

$V_L = I.X_L$ = voltage drop across L —leading I by π/2

$V_C = I.X_C$ = voltage drop across C —lagging I by π/2





$$\therefore \quad OD = \sqrt{OA^2 + AD^2} \quad \text{or} \quad V = \sqrt{(IR)^2 + (IX_L + IX_C)^2} = I\sqrt{R^2 + (X_L + X_C)^2}$$

$$\text{or} \quad I = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{V}{\sqrt{R^2 + X^2}} = \frac{V}{Z}$$

The term $\sqrt{R^2 + (X_L - X_C)^2}$ is known as the impedance of the circuit. Obviously,

$$(impedance)^2 = (resistance)^2 + (net\ reactance)^2$$

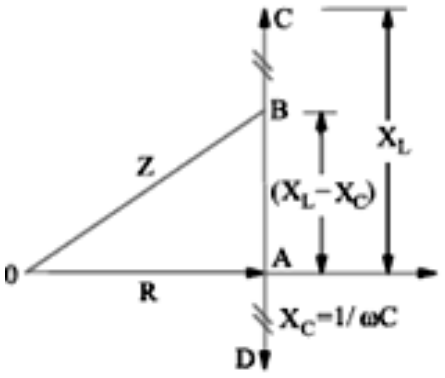
$$\text{or} \quad Z^2 = R^2 + (X_L - X_C)^2 = R^2 + X^2$$

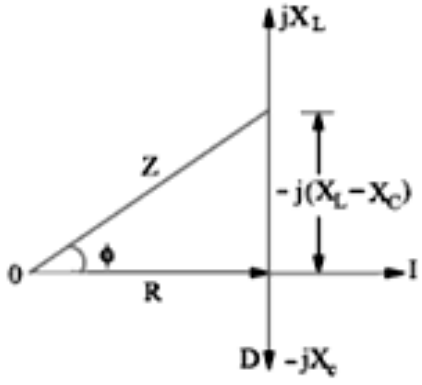
where

X is the net reactance,

Phase angle φ is given by $\tan \varphi = (X_L - X_C)/R = X/R$ = net reactance/resistance

$$\text{Power factor is } \cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{R}{\sqrt{R^2 + X^2}}$$





Case 1 : **$X_L = X_C$, circuit behaves like pure resistive circuit**

Case 2 : **$X_L > X_C$, circuit behaves like R-L circuit**

Case 3 : **$X_L < X_C$, circuit behaves like R-C circuit**

Hence, it is seen that if the equation of the applied voltage is $v = V_m \sin \omega t$, then equation of the resulting current in an *R-L-C* circuit is given by

$$\boldsymbol{i = I_m \sin(\omega t \pm \phi)}$$

where,

The + ve sign is to be used when current leads $\boldsymbol{X_C > X_L}$.

The − ve sign is to be used when current lags when $\boldsymbol{X_L > X_C}$.

In general, the current lags or leads the supply voltage by an angle ϕ such that tan ϕ = X/R
Using symbolic notation, we have

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Its phase angle is $\Phi = \tan^{-1}[X_L - X_C/R]$

$Z = Z \angle \tan^{-1}[(X_L - X_C)/R] = Z \angle \tan^{-1}(X/R)$

If $\mathbf{V} = V \angle 0$, then, $\mathbf{I} = \mathbf{V}/\mathbf{Z}$

The impedance in complex form is expressed as follows :

$$\mathbf{Z = R + j\,(X_L - X_C)}$$

Summary of Results of Series AC Circuits

Type of Impedance	*Value of Impedance*	*Phase angle for current*	*Power factor*
Resistance only	R	0°	1
Inductance only	ωL	90° lag	0
Capacitance only	$1/\omega C$	90° lead	0
Resistance and Inductance	$\sqrt{[R^2 + (\omega L)^2]}$	$0 < \phi < 90°$ lag	1 > p.f. > 0 lag
Resistance and Capacitance	$\sqrt{[R^2 + (-1/\omega C)^2]}$	$0 < \phi < 90°$ lead	1 > p.f. > 0 lead
R-L-C	$\sqrt{[R^2 + (\omega L \sim 1/\omega C)^2]}$	between 0° and 90° lag or lead	between 0 and unity lag or lead

Introduction To Three Phase AC Circuits or (Poly Phase AC Circuits)

Necessity and advantages of three phase systems

1. Three phase (3φ) power has a constant magnitude whereas single phase (1φ) power is pulsating.
2. A 3φ system can set up a rotating magnetic field whereas single phase supply is not so.
3. For the same rating 3φ machines are smaller, simpler in construction and have better operating characteristics than 1φ machines
4. weight of copper required for power transmission is less in three phase system when compared to single phase system.
5. Voltage regulation of a 3φ system is better than that of 1φ System .

Phase displacement: It is angular displacement between the phases.

Phase displacement between each phases for an n phase system $=360^0/n$,

where,

n = number of phases

Numbering (Naming) of the phases: A three phase supply (n = 3) is a set of three alternating quantities displaced from each other by an angle of 120^0 as shown below. The three phases may be named as R Y B, or A B C, or a b c or 1 2 3. In this analysis pases are named as **a, b** and **c**.

Generation of Three Phase Voltage

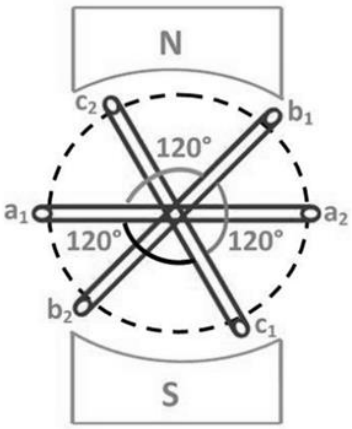


Fig : Three Phase AC Generator

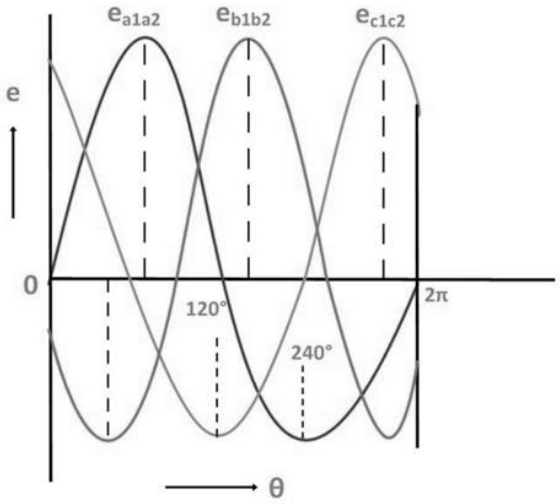


Fig : Three Phase AC Voltages

Three Phase voltage can be generated by placing three rectangular coils displaced in space by 120^0 in a uniform magnetic field as shown above. When these coils rotate in counter clockwise direction with a uniform angular velocity of ω rad/sec, a sinusoidal emf displaced by 120^0 is induced in these coils as shown below.

$$e_a = e_{a1a2} = E_m \sin \omega t$$

$$e_b = e_{b1b2} = E_m \sin(\omega t - 2\pi/3) = E_m \sin(\omega t - 120°)$$

$$e_c = e_{c1c2} = E_m \sin(\omega t - 4\pi/3) = E_m \sin(\omega t - 240°)$$

Phase 'a' waveform starts at 0^0. Phase 'b' waveform stars at 120^0 and phase 'c' waveform at 240^0.

Where

e_{a1a2}, e_{b1b2}, e_{c1c2} are the instantaneous value of the voltages in the phases a, b and c.

E_m is the maximum or peak value of the phase voltages

ω is the angular frequency in rad/secs

t is the time interval in seconds

The sum of the three phase voltages at any instant is equal to zero.

$$e_a + e_b + e_c = 0$$

Phase Sequence: The order in which the voltages in the three phases reach their maximum value is called as phase sequence. Here in the given waveform the phase sequence is ABC.

Phasor representation (Vector Diagram) of three phase Voltages:

The phasor representation of three phase voltages is shown below.

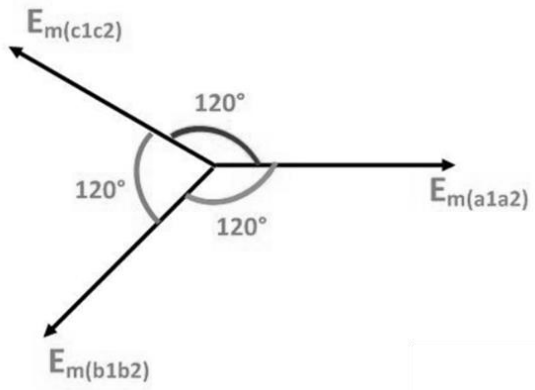


Fig : Vector diagram for representation of phase voltages with the phase sequence **abc**

The phase **a** voltage is taken as the reference and is drawn along the x-axis. The phase **b** voltage lags behind the phase **a** voltage by 120^0. The phase **c** voltage lags behind the phase **a** voltage by 240^0 and phase **b** voltage by 120^0.

The types of connections in three phase systems are as follows:

